

State Legislative Primary Election Returns Database, 1990–2026

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Abstract

State legislative primary elections play a central role in selecting lawmakers in the American states, yet the absence of a comprehensive national database has limited systematic research on their outcomes. We introduce a new database of state legislative primary election returns covering all 50 states and both legislative chambers from 1990 through 2026. This dataset contains 159,573 candidate-election observations spanning 90,860 primary races and records candidates' names, primary and primary-runoff vote totals, and incumbency statuses. To facilitate applied research, we also link this dataset to existing databases on candidate ideology, campaign finance filings, and general-election returns. This database will enable systematic research on candidate entry, primary competition, ideological extremism, electoral accountability, and legislative representation in the American states.

Keywords: election data, state legislatures, primary elections, American politics

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1 Background & Summary

State legislatures occupy a central place in American democracy, with nearly 7,400 legislators enacting policy in key areas including education, healthcare, criminal justice, and election administration. Primary elections are the first—and often decisive—stage in selecting these legislators. Yet despite their importance, state legislative primary elections have received far less scholarly attention than general elections or congressional primaries, in large part because comprehensive data on their outcomes has not been available.

In this paper, we introduce a new database of state legislative primary election returns covering all ninety-nine state legislative chambers for the years 1990–2026. Our database is the first comprehensive source of state legislative primary returns over the past three decades. It contains 159,573 candidate-election observations across 90,860 primary races involving approximately 95,645 distinct candidates. The database records each candidate’s vote total and win indicator, primary runoff vote totals and outcomes where applicable, candidate names, incumbency status, and district characteristics. We also link each candidate to Klarner’s state legislative election returns database (SLERs),¹ the Database on Ideology, Money in Politics, and Elections (DIME),² the National Institute on Money in State Politics (NIMSP),³ and Shor and McCarty’s legislator ideology (NP-score) estimates.^{4;5}

This new database will enable researchers to study a wide range of questions about primary elections, candidate selection, and legislative representation that have previously been difficult to address due to data limitations.

2 Methods

2.1 Data Collection

We constructed our database by collecting certified primary election returns for every state legislative primary election from 1990 through 2026. Our primary source for each state-chamber-year was the official Secretary of State or state election commission website, from which we downloaded machine-readable returns whenever available. When official digital returns were unavailable—most commonly for earlier years—we obtained data from state Blue Books and legislative manuals, or submitted direct requests to state election offices.

The resulting raw data varied substantially in format across states and years, including structured tabular files (CSV, JSON, and Excel), scanned PDFs, images of official canvass documents, and printed volumes. We developed custom parsing scripts to extract and standardize data from each format. Where automated extraction was not feasible, the data was

Table 1 – District Naming Conventions Used in American State Legislatures as of 2024. This table reports how states label their legislative districts. **Numeric** indicates numbers, **Alpha** indicates alphabetic characters, and **Post** indicates a position letter. U and L stand for upper and lower chambers, respectively.

Numeric	Alpha	Post	State-chambers
✓	—	—	AK (L), AL, AR, AZ, CA, CO, CT, DE, FL, GA, HI, IA, ID (U), IL, IN, KS, KY, LA, MD (U), ME, MI, MN (U), MO, MS, MT, NC, ND (L) [†] , ND (U), NE, NH (U), NJ, NM, NV, NY, OH, OK, OR, PA, RI, SC, SD, TN, TX, UT, VA, WA (U), WI, WV, WY
✓	✓	—	MA, NH (L), VT (L)
—	✓	—	AK (U), VT (U)
✓	—	✓	ID (L), MD (L), MN (L), WA (L)

[†]Districts 4 and 9 introduced candidate filing by position beginning in 2022.

entered manually. In a small number of cases where official sources could not be located, we supplemented our data with secondary sources (e.g., newspaper reports).

2.2 Standardization

Candidate names. The formatting of candidates’ names varies widely in our raw data across states and years. We standardized all names by converting them to uppercase; removing extraneous text such as titles (e.g., “Representative,” “Senator”), winner asterisks, and parenthetical notes; and stripping special characters. We then extracted first, last, middle, and suffix name components from the standardized name string using the Stata name-parsing routine `extrname`.

District identifiers. States use a variety of naming conventions to identify legislative districts. We standardized district identifiers by decomposing each into up to three components, provided as separate variables: `district_num` (the district number), `district_alpha` (an alphabetic name component, e.g., a county name in Massachusetts or a town name in Vermont), and `district_post` (a position designator used when candidates file for a specific seat within a multi-member district). Table 1 documents the resulting conventions by state-chamber.

Table 2 – Match Rates to External Datasets. Each row reports the share of candidate-election observations linked to an external database, using the denominator in its panel heading. DIME stands for Database on Ideology, Money in Politics, and Elections.² Follow the Money (FTM) provides state-level campaign finance disclosures from the National Institute on Money in State Politics.³ SLERs stand for State Legislative Election Returns.¹ NP-scores are the Shor-McCarty common-space legislator ideology estimates.⁵

Dataset	Content	Share matched
All primary candidates, 1990–2026 (N = 159,571)		
DIME	Ideology estimates	56.6%
Follow the Money	Campaign contributions	65.5%
Primary nominees, 1990–2024 (N = 83,694)		
Klarner SLERs	General-election returns	96.3%
Elected legislators, 1992–2021 (N = 50,424)		
Shor-McCarty	Legislator ideology (NP-scores)	91.1%

The post component required careful treatment in multi-member districts. In some chambers, candidates file for—and compete within—a specific post within a multi-member district (Idaho, Maryland, Minnesota, and Washington House). In these chambers the post variable defines a separate filing constituency (or race). In other multi-member chambers (North and South Dakota House and West Virginia Senate), candidates compete at-large within the district, with the top- k finishers winning k seats; since these positions are assigned post hoc and do not define separate races, we drop the post identifier for these races.

2.3 Linking to External Datasets

To facilitate future research, we link our primary-election database to four existing datasets on legislative politics. We use the Stata package `megamerge` to conduct fuzzy name matching within state-chamber-district-year blocks. Table 2 summarizes the four external datasets and our match rates. The table separates all primary candidates, primary nominees, and elected legislators to account for differences in the populations covered by these sources.

We first link to Klarner’s State Legislative Election Returns database,¹ which provides the corresponding general-election returns. This data covers general elections between 1990 and 2024. Matched candidates receive a `klarner_id` that can be used to pull any variable from the SLERs. We also incorporate the redistricting regime (`regime`) and single-member district indicator (`smd`) from the SLERs. Second, we match to the Database on Ideology,

Money in Politics, and Elections (DIME),² linking each matched candidate to a `bonica rid`. Third, we merge in identifiers from the National Institute on Money in State Politics (Follow the Money), which provides state-level campaign finance disclosures. Linked candidates receive an `ftm id`. DIME and FTM cover legislative candidates through the 2022 election cycle, with varying start dates around 2000. Finally, we merge in identifiers (`sm id`) from Shor and McCarty’s common-space ideology estimates (NP-scores). Shor-McCarty scores cover legislative service from 1993 through 2022, corresponding to election years 1992–2021 in our merged data.^{4;5}

3 Data Records

The database is available in three formats at <https://github.com/AndyCWMyers/state-legislative-primary-returns>: a Stata binary file (`.dta`), a comma-separated values file (`.csv`), and an R data file (`.rds`). All three files contain identical data: 159,573 rows and 26 variables. Each row represents an individual candidate in a single primary election. Table 3 describes the variables.

Table 3 – Variables Included in Database. Coverage reports the first and last election years with nonmissing values; availability varies by state, year, and candidate.

Variable	Description	Coverage
Identifiers		
year	Election year.	1990–2026
state	State (2-letter USPS abbreviation).	1990–2026
chamber	Legislative chamber (U = upper/senate, L = lower/house).	1990–2026
party	Party (dem = Democratic, rep = Republican, NP = Nebraska nonpartisan).	1990–2026
district_num	District number component (missing if absent).	1990–2026
district_alpha	Named district component.	1990–2026
district_post	Within-district position designator.	1990–2026
primary_race_id	Unique primary race ID (state $\times$ year $\times$ chamber $\times$ district $\times$ party).	1990–2026
klarner_id	Klarner Politics general election data candidate ID.	1990–2024
bonica rid	DIME recipient ID. ²	1992–2022
ftm_id	Follow the Money / NIMSP candidate ID.	1992–2022
sm_id	Shor-McCarty legislator ID for NP-score ideal points. ^{4,5}	1992–2021
Candidate name		
first_name	Candidate first name (standardized).	1990–2026
middle_name	Candidate middle name (standardized).	1990–2026
last_name	Candidate last name (standardized).	1990–2026
name_suffix	Candidate name suffix (Jr., Sr., III, etc.).	1990–2026
Primary election		
votes_primary	Candidate’s raw vote total in the primary election.	1990–2026
win_primary	=1 if the candidate received the highest primary vote total.	1990–2026
Primary runoff		
ran_primary_runoff	=1 if the candidate participated in a primary runoff.	1990–2026
votes_primary_runoff	Candidate’s raw vote total in the primary runoff.	1990–2026
win_primary_runoff	=1 if the candidate won the primary runoff.	1990–2026
win_primary_final	Uses win_primary_runoff if a runoff occurs; otherwise win_primary .	1990–2026
Incumbency		
inc	=1 if the candidate held a seat in the same chamber at the time of the election.	1990–2024
District characteristics		
regime	Redistricting regime (increments with each post-Census redistricting cycle).	1990–2026
smd	=1 if the election is in a single-member district.	1990–2026
district_seats	Number of seats available in the district.	1990–2026

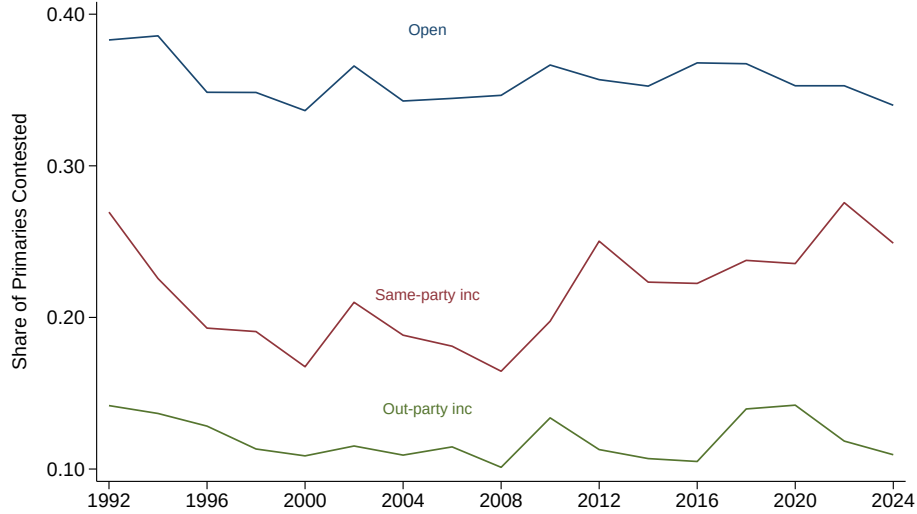
4 Data Overview

Table 4 summarizes the composition of the database by legislative chamber, incumbency, party, election timing, and runoff participation. Approximately four in five candidate-election observations are from lower chambers, three in ten correspond to incumbents, and Democratic and Republican candidates are represented in roughly equal proportions. The database also makes it possible to describe primary competition over time and across electoral contexts. Figure 1 summarizes the share of contested primaries in even-year elections from 1992 through 2024, distinguishing open-seat races, races featuring a same-party incumbent, and races in districts held by an opposing-party incumbent. As the figure shows, the majority of state legislative primaries are uncontested, with modest upticks in competition in major redistricting years.

Table 4 – Dataset Composition Across Candidate- and Race-Level Characteristics.

	Observations	Share
<i>Total</i>		
All candidate-elections	159,573	100.0%
<i>Chamber</i>		
House	127,828	80.1%
Senate	31,745	19.9%
<i>Incumbency</i>		
Incumbents	44,377	27.8%
Non-incumbents	102,089	64.0%
Missing incumbency	13,105	8.2%
<i>Party</i>		
Democratic	80,347	50.4%
Republican	78,522	49.2%
Nonpartisan	704	0.4%
<i>Election timing</i>		
Even-year	150,403	94.3%
Odd-year	9,170	5.7%
<i>Runoffs</i>		
No primary runoff	157,082	98.4%
Participated in primary runoff	2,489	1.6%

Figure 1 – Share of Contested Primaries by Electoral Context. Each line shows the average share of primary elections that are contested in even years, stratified across electoral environments. Open races are primary elections without an incumbent, same-party incumbent races feature an incumbent from the party holding the primary, and out-party primaries occur in districts held by an incumbent from the opposing party.



5 Technical Validation

5.1 Randomized Spot-Checks

After our collection was complete, the data underwent two systematic checks. First, research assistants verified the completeness of contested primary election records by cross-referencing the collected data against a comprehensive list of expected contested races. For the portion of the database spanning 1990 to 2016, we deliberately withheld at least four contested races from each state’s checklist that research assistants were required to identify and add. Second, research assistants verified the accuracy of candidate names and vote totals for a random sample of at least 15% (1990–2016) and 5% (2017–2024) of races per state.

5.2 Internal Consistency Checks

We also performed several internal consistency checks on our primary-election vote data.

Win indicator consistency. We verified that all candidates with `win_primary = 1` have the highest recorded vote total in their primary race. The win indicators do not separately

identify all nominees in multi-member districts or all candidates advancing in systems that advance multiple candidates.

Runoff consistency. For state-years that use a primary runoff system, we verified that runoff variables are populated for the appropriate candidates.

6 Usage Notes

Three features of our dataset warrant special attention. First, each row represents a candidate-election. Since many states do not publish vote totals for uncontested primaries, however, candidates winning uncontested primary elections often do not appear in our database. We include data on uncontested primary elections whenever available, but researchers wanting to recover the identity of winners of uncontested primary elections should merge our database to Klarner’s SLERs.

Second, Louisiana uses a jungle primary in which all candidates compete on a single ballot regardless of party, with a runoff between the top two finishers if no candidate receives a majority. We treat the first-stage contest as the primary election and include it in the dataset. And finally, since Nebraska’s unicameral legislature is formally nonpartisan, we set these candidates’ party to NP (for no party).

7 Data Availability

Our database is available at <https://github.com/AndyCWMyers/state-legislative-primary-returns>.

8 Code Availability

The replication code for the analyses presented in this paper is available at <https://github.com/AndyCWMyers/state-legislative-primary-returns>.

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10 Author Contributions

A.M. conceived and designed the database, coordinated data collection, wrote the data cleaning and assembly pipeline, and wrote the manuscript. S.R. contributed to data collection and to writing the manuscript. A.F., A.H., C.H.N., and J.W. contributed to data collection.

11 Competing Interests

The authors declare no competing interests.

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